Turbulence Modeling Resource

The Spalart-Allmaras Turbulence Model

This web page gives detailed information on the equations for various forms of the Spalart-Allmaras turbulence model. All forms of the model given on this page (EXCEPT for SA-QCR2000, SA-QCR2013 / SA-QCR2013-V, SA-QCR2020, and SA-QCR2024) are linear eddy viscosity models. Linear models use the Boussinesq assumption for the constitutive relation:

$$\tau_{ij} = 2\mu_t \left(S_{ij} - \frac{1}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij} \right) - \frac{2}{3}\rho k \delta_{ij}$$

where the last term is generally ignored for one-equation models like this one because *k* is not readily available (the term is sometimes ignored for non-supersonic speed flows for other models as well). The SA-QCR2000 section below provides details regarding a nonlinear implementation. The SA-QCR2013 / SA-QCR2013-V, SA-QCR2020 and SA-QCR2024 sections below provides details regarding a nonlinear implementation that also includes an approximation for the -2/3 rho k del_ij term.

Unless otherwise stated, for compressible flow with heat transfer this model is implemented as described on the page Implementing Turbulence Models into the Compressible RANS Equations, with perfect gas assumed and Pr = 0.72, Pr_t = 0.90, and Sutherland's law for dynamic viscosity.

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OVERVIEW

General Model

- (SA)
- (SA-neg)
- (SA-noft2)
- (SA-Ia)

The first version listed (SA) is considered "standard". It is the original published version without the rarely-used primary trip term present in (SA-Ia). The version (SA-neg) should yield essentially identical results to (SA), and is generally recommended because of its more robust numerical behavior. The version (SA-noft2) is a commonly-used variant of (SA), and should yield essentially identical results for most problems of practical interest, since in (SA) the sufficiently large inflow values for the turbulence field variable over-rides the effect of the f_{t2} term. However, note that when using Spalart-Allmaras as the basis for Detached-Eddy Simulation (Proc. 1st AFOSR International Conference, ed. C. Liu and Z. Liu, Greyden Press, Columbus OH 1997, pp. 137-147) or DDES (Theor. Comput. Fluid Dyn. (2006) 20:181-195, https://doi.org/10.1007/s00162-006-0015-0), the f_{t2} term present in (SA) may cause an undesirable delay in transition to turbulence in the RANS region; the version (SA-noft2) can help in this situation (see Vatsa et al, AIAA J 55(8), 2017, pp. 2842-2847, https://doi.org/10.2514/1.J055685). The trip version (SA-Ia) is rarely used.

Corrections to the General Model

These corrections can be applied individually or together in combination with the General Model.

- (RC) or (R) or (KL)
- (LRe)
- (comp)
- (rough)
- (TC)
- (QCR2000) or (QCR2013 / QCR2013-V) or (QCR2020) or (QCR2024)
- (Helicity)

For example, one could implement various combinations like: SA-RC, SA-neg-RC, SA-noft2-RC, SA-noft2-R, SA-KL-LRe-QCR2013-V, SA-neg-RC-QCR2020, or even SA-neg-RC-LRe-comp-rough-TC-QCR2000-Helicity. Here, for clarity, the particular "general model" being used is underlined, and the "corrections" are given in red. (Note that one could not combine R and RC, or different

QCR variants, together in the same model implementation.)

Other Versions of SA

The following represent some of the parallel versions of SA that have come out in the literature; they are considered different models, and, unlike the corrections, are not compatible with the General Model.

- (SA-noft2-Catris)
- (SA-noft2-Edwards)
- (SA-fv3)
- (SA-noft2-salsa)

"Standard" Spalart-Allmaras One-Equation Model (SA)

The following equations represents the most commonly-used implementation of the Spalart-Allmaras model (written in non-conservation form). The primary reference is:

- Spalart, P. R. and Allmaras, S. R., "A One-Equation Turbulence Model for Aerodynamic Flows," **Recherche Aerospatiale**, No. 1, 1994, pp. 5-21. (paper provided with permission of authors)

This journal reference is the official publication that arose from the AIAA Conference Paper AIAA-92-0439 (Reno, NV, January 1992). However, there are some differences, and the journal reference takes precedence. For example, AIAA-92-0439 used c_{t3}=1.1 and c_{t4}=2.0; these constants are now different (although for fully turbulent solutions, these sets of constants should make negligible difference). Note that this journal reference had a small typo (appendix only) in its definition of the constant c_{w1} (missing square on the kappa term in the denominator). The typo, which sometimes gets propagated to other reports/papers, is corrected below. The original reference made use of a trip term that most people do not include, because the model is most often employed for fully turbulent applications. Therefore, in this "standard" representation the trip term is being left out (see version (SA-Ia) below for the version including the trip term). As a consequence, the farfield boundary condition must be changed from that given in the above reference. The new farfield boundary condition is taken from the following references:

- Spalart, P. R., "Trends in Turbulence Treatments," **AIAA 2000-2306**, June 2000, https://doi.org/10.2514/6.2000-2306.
- Spalart, P. R. and Rumsey, C. L., "Effective Inflow Conditions for Turbulence Models in Aerodynamic Calculations," **AIAA Journal**, Vol. 45, No. 10, 2007, pp. 2544-2553, https://doi.org/10.2514/1.29373.

In all of the following, a "hat" is used over the turbulence field variable, rather than a "tilde" as given in the references, for the sole practical reason that the "tilde" showed up very poorly on the screen.

The one-equation model is given by the following equation:

$$\frac{\partial \hat{\nu}}{\partial t}+u_j\frac{\partial \hat{\nu}}{\partial x_j} = c_{b1}(1-f_{t2})\hat{S}\hat{\nu}-\left[c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2}\right]\left(\frac{\hat{\nu}}{d}\right)^2+\frac{1}{\sigma}\left[\frac{\partial}{\partial x_j}\left((\nu+\hat{\nu})\frac{\partial \hat{\nu}}{\partial x_j}\right)+c_{b2}\frac{\partial \hat{\nu}}{\partial x_i}\frac{\partial \hat{\nu}}{\partial x_i}\right]$$

and the turbulent eddy viscosity is computed from:

$$\mu_t = \rho\hat{\nu}f_{v1}$$

where

$$f_{v1} = \frac{\chi^3}{\chi^3+c_{v1}^3}$$
$$\chi = \frac{\hat{\nu}}{\nu}$$

and ρ is the density, $\nu = \mu/\rho$ is the molecular kinematic viscosity, and μ is the molecular dynamic viscosity. Additional definitions are given by the following equations:

$$\hat{S} = \Omega + \frac{\hat{\nu}}{\kappa^2 d^2}f_{v2}$$

where $\Omega = \sqrt{2W_{ij}W_{ij}}$ is the magnitude of the vorticity, *d* is the distance from the field point to the nearest wall, and

$$f_{v2} = 1-\frac{\chi}{1+\chi f_{v1}} \qquad f_w = g\left[\frac{1+c_{w3}^6}{g^6+c_{w3}^6}\right]^{1/6}$$

$$g = r + c_{w2}(r^6 - r)$$

$$r = min\left[\frac{\hat{\nu}}{\hat{S}\kappa^2 d^2}, 10\right]$$

$$f_{t2} = c_{t3}exp\left(-c_{t4}\chi^2\right)$$

$$W_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right)$$

The boundary conditions are:

$$\hat{\nu}_{wall} = 0 \qquad \hat{\nu}_{farfield} = 3\nu_\infty : to : 5\nu_\infty$$

Note that these boundary conditions on the SA turbulence field variable correspond to turbulent kinematic viscosity values of:

$$\nu_{t,wall} = 0 \qquad \nu_{t,farfield} = 0.210438\nu_\infty : to : 1.294234\nu_\infty$$

The constants are:

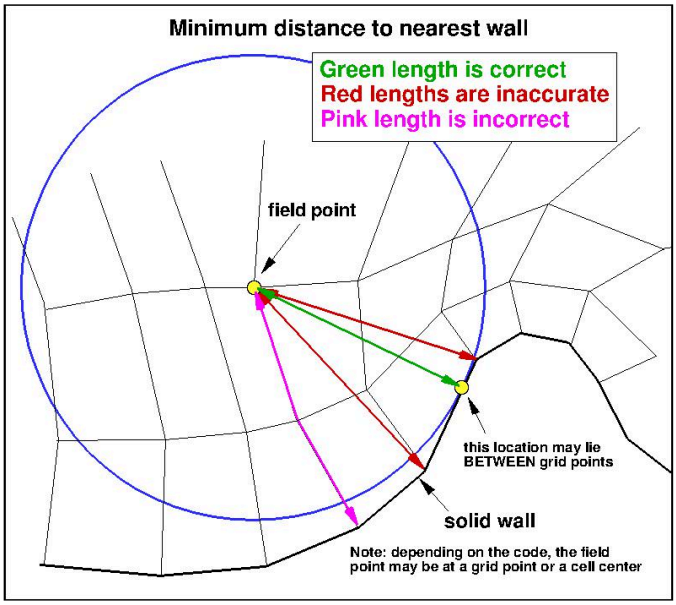
$$c_{b1} = 0.1355 \qquad \sigma = 2/3 \qquad c_{b2} = 0.622 \qquad \kappa = 0.41$$

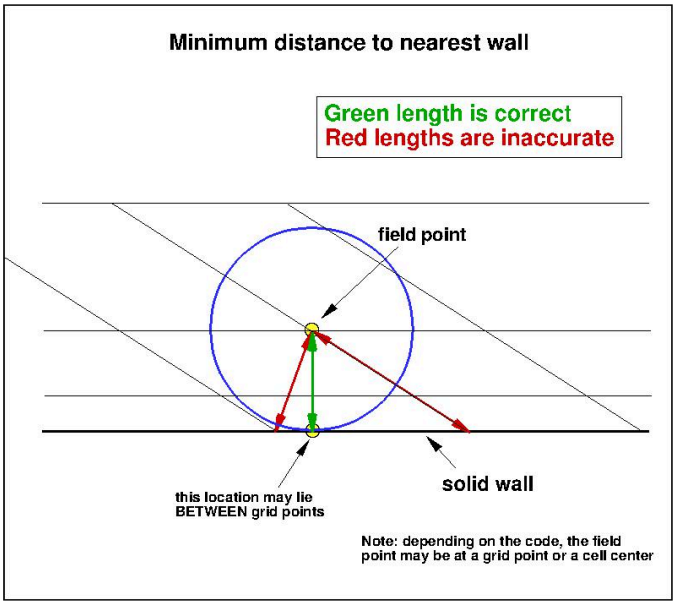
$$c_{w2} = 0.3 \qquad c_{w3} = 2 \qquad c_{v1} = 7.1 \qquad c_{t3} = 1.2 \qquad c_{t4} = 0.5$$

$$c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1 + c_{b2}}{\sigma}$$

Note that this model has source terms (production and destruction) that are non-zero in the freestream, even when vorticity is zero. The source terms are, however, very small: proportional to $1/d^2$.

It is important to note that computing minimum distance (*d*) by searching along grid lines or by finding the nearest wall gridpoint (or cell center) are incorrect, and are **not** the same as computing actual minimum distance to the nearest wall (in general). Using the former methods will yield grid-dependent differences in the results. The following sketches demonstrate the concept of minimum distance. Improperly-calculated minimum distance functions will particularly produce incorrect results for cases in which the grid lines are not perfectly normal to the body surface, or when the nearest body does not lie in the current grid zone. Note that when the nearest wall point is a sharp convex corner or edge (like an airfoil or wing trailing edge) then the correct minimum distance is the distance to that corner or edge, which is not a wall normal.





<u>Note 1</u>: To avoid possible numerical problems, the term $\hat{S}$ when it is supplied to the calculation of *r* must never be allowed to reach zero or go negative.

- (a) One common method limits the term to be greater than zero.
- (b) Spalart (private communication) recommends limiting $\hat{S}$ to be no smaller than 0.3*Ω.
- (c) Another method (that avoids clipping for non-zero Ω) is described in Allmaras, S. R., Johnson, F. T., and Spalart, P. R., "Modifications and Clarifications for the Implementation of the Spalart-Allmaras Turbulence Model," ICCFD7-1902, 7th International Conference on Computational Fluid Dynamics, Big Island, Hawaii, 9-13 July 2012 (https://www.iccfd.org/iccfd7/assets/pdf/papers/ICCFD7-1902_paper.pdf). The equation for $\hat{S}$ (above) is replaced. First, define:

  $$\overline{S} = \frac{\hat{\nu}}{\kappa^2 d^2} f_{v2}$$

  Then:

  $$\hat{S} = \Omega + \overline{S} \quad \text{when} \quad \overline{S} \geq -c_2\Omega$$

  $$\hat{S} = \Omega + \frac{\Omega(c_2^2\Omega + c_3\overline{S})}{(c_3 - 2c_2)\Omega - \overline{S}} \quad \text{when} \quad \overline{S} < -c_2\Omega$$

  where $c_2 = 0.7$ and $c_3 = 0.9$. Note that $\hat{S}$ may be zero in certain situations when the vorticity magnitude is identically zero. As a result, there needs to be a guard on the computation of *r*, to avoid divide-by-zero: whenever $\hat{S}$ = 0, set *r* = 10.

  The particular limiting method used should always be reported.

Note 2: Allmaras, S. R., "Multigrid for the 2-D Compressible Navier-Stokes Equations," AIAA Paper 99-3336, June-July 1999, https://doi.org/10.2514/6.1999-3336 discusses two optional modifications to the model that have no effect on a non-negative (i.e., meaningful) steady-state turbulence solution, but are designed to help robustness for cases where grid stretching is excessive, grid smoothness poor, or grid resolution inadequate. These modifications were intended to aid solution of the original equations, and **not** to be a new model correction. See the (SA-neg) description below for updated modifications. Other modifications along the same lines also exist in the literature; see, for example: Crivellini et al (J. Computational Physics 241:388-415, 2013).

Note 3: Spalart and Allmaras recommend the use of the following "turbulence index" i_t at walls to detect transition:

$$i_t = \frac{1}{\kappa u_\tau}\frac{\partial \hat{\nu}}{\partial n}$$

where *n* is the wall-normal direction, and one can approximate u_τ with: $u_\tau \approx \sqrt{\nu\Omega}$. The index will be close to zero for a laminar region and close to 1 for a turbulent region.

Note 4: Some implementations of SA have been noted to incorrectly introduce the density inside the derivatives of the SA equation (in contrast with the Catris-Aupoix proposal, which has a valid basis). This will alter the SA predictions for supersonic flows. Users need to be aware of the equation details in the solver they are using.

Negative Spalart-Allmaras One-Equation Model (SA-neg)

This modification to the Spalart-Allmaras model was developed primarily to address issues with under-resolved grids and non-physical transient states in discrete settings. It was formulated to "be passive to the original (SA) model in well-resolved flowfields and should produce negligible differences in most cases." The reference is:

- Allmaras, S. R., Johnson, F. T., and Spalart, P. R., "Modifications and Clarifications for the Implementation of the Spalart-Allmaras Turbulence Model," ICCFD7-1902, 7th International Conference on Computational Fluid Dynamics, Big Island, Hawaii, 9-13 July 2012 (https://www.iccfd.org/iccfd7/assets/pdf/papers/ICCFD7-1902_paper.pdf).

The model is the same as the "standard" version (SA) when the turbulence variable $\hat{\nu}$ is greater than or equal to zero; this includes Note 1 concerning issues with $\hat{S}$ and *r* (use of Note 1 (c) is considered standard for this model). When $\hat{\nu}$ is negative the following equation is solved instead:

$$\frac{\partial \hat{\nu}}{\partial t}+u_j\frac{\partial \hat{\nu}}{\partial x_j} = c_{b1}(1-c_{t3})\Omega\hat{\nu}+c_{w1}\left(\frac{\hat{\nu}}{d}\right)^2+\frac{1}{\sigma}\left[\frac{\partial}{\partial x_j}\left((\nu+\hat{\nu}f_n)\frac{\partial \hat{\nu}}{\partial x_j}\right)+c_{b2}\frac{\partial \hat{\nu}}{\partial x_i}\frac{\partial \hat{\nu}}{\partial x_i}\right]$$

with

$$f_n = \frac{c_{n1}+\chi^3}{c_{n1}-\chi^3}$$

and $c_{n1} = 16$.

The turbulent eddy viscosity (μ_t) in the momentum and energy equations is set to zero when $\hat{\nu}$ is negative.

Note that the sign of the destruction term $c_{w1}(\hat{\nu}/d)^2$ is "+" as written; this is opposite of the positive model. All other constants and variables are the same as defined for the "standard" version (SA).

The above paper also describes other important information regarding the SA model. For example, the authors reaffirm that the original SA model, in which density does not appear, is applicable to both incompressible and compressible flows, and it should be considered the standard form for compressible. However, they show that an equivalent conservation form can be constructed by combining SA with the mass conservation equation, yielding (for standard SA):

$$\frac{\partial(\rho\hat{\nu})}{\partial t}+\frac{\partial(\rho u_j\hat{\nu})}{\partial x_j} = \rho c_{b1}(1-f_{t2})\hat{S}\hat{\nu}-\rho\left[c_{w1}f_w-\frac{c_{b1}}{\kappa^2}f_{t2}\right]\left(\frac{\hat{\nu}}{d}\right)^2+\frac{1}{\sigma}\left[\frac{\partial}{\partial x_j}\left(\rho(\nu+\hat{\nu})\frac{\partial \hat{\nu}}{\partial x_j}\right)+\rho c_{b2}\frac{\partial \hat{\nu}}{\partial x_i}\frac{\partial \hat{\nu}}{\partial x_i}\right]-\frac{1}{\sigma}(\nu+\hat{\nu})\frac{\partial \rho}{\partial x_i}\frac{\partial \hat{\nu}}{\partial x_i}$$

Furthermore, the authors state that there should never be hard-wired limits on the maximum value of eddy viscosity produced by the model. In attached boundary layers, the maximum value of χ across the profile grows with streamwise location like $\chi_{max} \approx 0.00059Re_x^{0.83}$. Asymptotic values in wakes and jets are independent of streamwise distance; these particular relations can be found in the above-mentioned paper.

Spalart-Allmaras One-Equation Model without f_{t2} Term (SA-noft2)

Many implementations of Spalart-Allmaras ignore the f_{t2} term, which was a numerical fix in the original model in order to make zero a stable solution to the equation with a small basin of attraction (thus slightly delaying transition so that the trip term could be activated appropriately). It is argued that if the trip is not used, then f_{t2} is not necessary. The equations are the same as for the "standard" version (SA), except that the term f_{t2} does not appear at all (i.e., c_{t3}=0 instead of 1.2). Two examples of references that use this form are:

- Eca, L., Hoekstra, M., Hay, A., and Pelletier, D., "A Manufactured Solution for a Two-Dimensional Steady Wall-Bounded Incompressible Turbulent Flow," **International Journal of Computational Fluid Dynamics**, Vol. 21, Nos. 3-4, 2007, pp. 175-

188, https://doi.org/10.1080/10618560701553436.

- Aupoix, B. and Spalart, P. R., "Extensions of the Spalart-Allmaras Turbulence Model to Account for Wall Roughness," **International Journal of Heat and Fluid Flow**, Vol. 24, No. 4, 2003, pp. 454-462, https://doi.org/10.1016/S0142-727X(03)00043-2.

Based on studies (see, e.g., Rumsey, C. L., "Apparent Transition Behavior of Widely-Used Turbulence Models," **International Journal of Heat and Fluid Flow**, Vol. 28, 2007, pp. 1460-1471, https://doi.org/10.1016/j.ijheatfluidflow.2007.04.003), use of this form as opposed to the "standard" version (SA) probably makes very little difference, at least at reasonably high Reynolds numbers, provided that the "standard" version use the appropriate boundary condition of $\hat{\nu}_{farfield} = 3\nu_\infty$ (or greater).

Spalart-Allmaras One-Equation Model with Trip Term (SA-Ia)

The form of the Spalart-Allmaras model with the trip term included is given in the following reference:

- Spalart, P. R. and Allmaras, S. R., "A One-Equation Turbulence Model for Aerodynamic Flows," **Recherche Aerospatiale**, No. 1, 1994, pp. 5-21. (paper provided with permission of authors)

The equations are the same as for the "standard" version (SA), except there is an additional trip term on the right hand side of the equation:

$$\frac{\partial \hat{\nu}}{\partial t} + u_j \frac{\partial \hat{\nu}}{\partial x_j} = c_{b1}(1-f_{t2})\hat{S}\hat{\nu} - \left[c_{w1} f_w - \frac{c_{b1}}{\kappa^2} f_{t2}\right]\left(\frac{\hat{\nu}}{d}\right)^2 + \frac{1}{\sigma}\left[\frac{\partial}{\partial x_j}\left((\nu+\hat{\nu})\frac{\partial \hat{\nu}}{\partial x_j}\right) + c_{b2}\frac{\partial \hat{\nu}}{\partial x_i}\frac{\partial \hat{\nu}}{\partial x_i}\right] + f_{t1}\Delta U^2$$

where:

$$f_{t1} = c_{t1} g_t exp\left[-c_{t2}\frac{\omega_t^2}{\Delta U^2}(d^2 + g_t^2 d_t^2)\right]$$

$$g_t = min\left[0.1, \frac{\Delta U}{\omega_t \Delta x_t}\right]$$

and ΔU is the difference between the velocity at the field point and that at the trip (on the wall), Δx_t is the grid spacing along the wall at the trip, ω_t is the wall vorticity at the trip, d_t is the distance from the field point to the trip, $c_{t1} = 1$, and $c_{t2} = 2$.

The farfield boundary condition is:

$$0 \le \hat{\nu}_{farfield} < \frac{1}{10}\nu_\infty$$

Note that this boundary condition on the SA turbulence field variable corresponds to turbulent kinematic viscosity values of:

$$0 \le \nu_{t,farfield} < 0.27940 \times 10^{-6}\nu_\infty$$

Spalart-Allmaras One-Equation Model with Rotation/Curvature Correction (SA-RC)

This form of the Spalart-Allmaras model attempts to account for rotation and curvature effects. The reference is:

- Shur, M. L., Strelets, M. K., Travin, A. K., Spalart, P. R., "Turbulence Modeling in Rotating and Curved Channels: Assessing the Spalart-Shur Correction," **AIAA Journal** Vol. 38, No. 5, 2000, pp. 784-792, https://doi.org/10.2514/2.1058.

An earlier reference (Spalart & Shur, **Aerospace Science and Technology** 5:297-302, 1997) has typographical errors, but contains a useful physical discussion of the RC term. The model is the same as for the "standard" version (SA), except that the $c_{b1}\hat{S}\hat{\nu}$ term gets multiplied by the rotation function f_{r1}:

$$f_{r1} = (1+c_{r1})\frac{2r^*}{1+r^*}\left[1 - c_{r3} tan^{-1}(c_{r2}\hat{r})\right] - c_{r1}$$

Specifically, the first term on the RHS of the SA equation becomes: $c_{b1}(f_{r1} - f_{t2})\hat{S}\hat{\nu}$. The various terms are:

$$r^* = S/\omega$$

$$\hat{r} = \frac{2\omega_{ik}S_{jk}}{D^4}\left(\frac{DS_{ij}}{Dt} + (\varepsilon_{imn}S_{jn} + \varepsilon_{jmn}S_{in})\Omega'_m\right)$$

$$S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \qquad \omega_{ij} = \frac{1}{2}\left[\left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i}\right) + 2\varepsilon_{mji}\Omega'_m\right]$$

$$S^2 = 2S_{ij}S_{ij} \qquad \omega^2 = 2\omega_{ij}\omega_{ij}$$

$$D^2 = \frac{1}{2}(S^2 + \omega^2)$$

$$c_{r1} = 1.0 \qquad c_{r2} = 12 \qquad c_{r3} = 1.0$$

The term DS_{ij}/Dt represents the components of the Lagrangian derivative of the strain rate tensor. The rotation rate Ω' is used only if the reference frame itself is rotating (note that all derivatives should be defined with respect to the reference frame). Note that the Lagrangian (or material) derivative is:

$$\frac{DS_{ij}}{Dt} \equiv \frac{\partial S_{ij}}{\partial t} + u_k \frac{\partial S_{ij}}{\partial x_k}$$

It is not unusual for RC implementations to ignore the time term, but this is only correct for steady-state problems in which the time rate of change of the flowfield is zero. Without the time term, the model is not correct in a time-dependent flowfield. This is especially relevant to helicopter rotor simulations.

Note that if this model is applied to the (SA-noft2) version instead, then its naming convention becomes (SA-noft2-RC). *Note also that the SA-RC production term can be negative.*

Spalart-Allmaras One-Equation Model with Rotation Correction (SA-R)

This correction to the SA model reduces the eddy viscosity in regions where vorticity exceeds strain rate, such as in vortex core regions where pure rotation should not produce turbulence, and should in fact suppress it according to some theories. The modification should be passive in thin shear layers where vorticity and strain are very close. This model can be viewed as a less capable but far simpler alternate to SA-RC. Two reference for this modification are:

- Dacles-Mariani, J., Zilliac, G. G., Chow, J. S., and Bradshaw, P., "Numerical/Experimental Study of a Wingtip Vortex in the Near Field," **AIAA Journal**, Vol. 33, No. 9, 1995, pp. 1561-1568, https://doi.org/10.2514/3.12826.
- Dacles-Mariani, J., Kwak, D., and Zilliac, G. G., "On Numerical Errors and Turbulence Modeling in Tip Vortex Flow Prediction," **Int. J. for Numerical Methods in Fluids**, Vol. 30, 1999, pp. 65-82, https://doi.org/10.1002/(SICI)1097-0363(19990515)30:1<65::AID-FLD839>3.0.CO;2-Y.

This model is the same as the "standard" version (SA), except that the (SA-R) production term becomes:

$$c_{b1}(1 - f_{t2})[\hat{S} + C_{rot} min(0, S - \Omega)]\hat{\nu}$$

where $S = \sqrt{2S_{ij}S_{ij}}$, and

$$S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

The constant C_{rot} represents an attempt to empirically adjust the production term for vortex dominated flows. The value of C_{rot}=2 was recommended in the above references. Note that when C_{rot} is greater than unity, the production term can be negative. When this occurs, the model is suppressing eddy viscosity, which is believed to be correct in solid-body rotation.

However, a negative production term is not acceptable for the negative branch ($\hat{\nu} < 0$) of the (SA-neg) model. If coding (SA-neg-R), a possible C0 continuous production term in the negative (SA-neg-R) branch is the following:

$$c_{b1}(1 - c_{t3}) abs[\Omega + C_{rot} min(0, S - \Omega)]\hat{\nu}$$

Note that if this model is applied to the (SA-noft2) version instead, then its naming convention becomes (SA-noft2-R). Also note that if a different value for C_{rot} is employed, it should be indicated as well. For example, when using a value of 1 instead of 2, the (SA-noft2-R) would instead have the naming convention (SA-noft2-R(C_{rot}=1)). (See, for example, AIAA Paper 2022-3743, June-July 2022, https://doi.org/10.2514/6.2022-3743.)

Note that (SA-noft2) is not compatible with (SA-neg) (see ICCFD7-1902, https://www.iccfd.org/iccfd7/assets/pdf/papers/ICCFD7-1902_paper.pdf), so (SA-neg-noft2-R) is currently not viable without additional modifications.

Spalart-Allmaras One-Equation Model with Kato-Launder Correction (SA-KL)

This correction to the SA model reduces the eddy viscosity in regions where vorticity exceeds strain rate, such as in vortex core regions where pure rotation should not produce turbulence. The modification should be passive in thin shear layers where vorticity and strain are very close. Like SA-R, this model can be viewed as a less capable but far simpler alternate to SA-RC. Two reference for this modification are:

- Kato, M. and Launder, B. E., "The Modelling of Turbulent Flow Around Stationary and Vibrating Square Cylinders," 9th Symposium on Turbulent Shear Flows, Kyoto, Japan, August 1993, paper 10-4, https://www.researchgate.net/publication/247931894_The_Modelling_of_Turbulent_Flow_Around_Stationary_and_Vibrating_Squ
- Nichols, R. H., "A Summary of the Turbulence Models in the CREATE™-AV Kestrel Flow Solvers," AIAA Paper 2019-1342, January 2019, https://doi.org/10.2514/6.2019-1342.

This model is the same as the "standard" version (SA), except that the magnitude of vorticity Ω (in the production term only) gets replaced by $\sqrt{S\Omega}$

where $S = \sqrt{2S_{ij}S_{ij}}$, and

$$S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

In other words, the Ω in $\hat{S}$ gets replaced only where $\hat{S}$ appears in the production term, and *not* where $\hat{S}$ appears in the *r* term.

Spalart-Allmaras Low Reynolds Number Version (SA-LRe)

This correction improves SA behavior for low Reynolds numbers, based on boundary-layer thickness. The reference is:

- Spalart, P. R. and Garbaruk, A. V., "Correction to the Spalart-Allmaras Turbulence Model, Providing More Accurate Skin Friction," **AIAA Journal**, Vol. 58, No. 5, May 2020, pp. 1903-1905, https://doi.org/10.2514/1.J059489.

The model is identical to the original model, except that the constant c_{w2} is changed to the following function:

$$c_{w2,LRe} = c_{w4} + \frac{c_{w5}}{((\chi/40)+1)^2}$$

with $c_{w4} = 0.21$ and $c_{w5} = 1.5$.

Mixing Layer Compressibility Correction in Spalart-Allmaras One-Equation Model (SA-comp)

This correction improves SA behavior in compressible mixing layers. The reference is:

- Spalart, P. R., "Trends in Turbulence Treatments," **AIAA 2000-2306**, June 2000, https://doi.org/10.2514/6.2000-2306.

This version is the same as for the "standard" version (SA), except that the following additional term is included on the right hand side of the equation.

$$-C_5 \frac{\hat{\nu}^2}{a^2}\frac{\partial u_i}{\partial x_j}\frac{\partial u_i}{\partial x_j}$$

where *a* is the local speed of sound and $C_5 = 3.5$.

Note that this correction is based on work in Shur, M., Strelets, M., Zaikov, L., Gulyaev, A., Kozlov, V., and Secundov, A., "Comparative Numerical Testing of One- and Two-Equation Turbulence Models for Flows with Separation and Reattachment," AIAA 95-0863, January 1995, https://doi.org/10.2514/6.1995-863. However, the form of the correction is somewhat different.

If used in conjunction with SA-noft2 instead (see for example Forsythe, J. R., Hoffmann, K. A., Squires, K. D., AIAA 2002-0586, 2002, https://doi.org/10.2514/6.2002-586), the model name would become SA-noft2-comp.

Wall Roughness Correction in Spalart-Allmaras One-Equation Model (SA-rough)

This correction gives SA capability for predicting rough walls. The references are:

- Aupoix, B. and Spalart, P. R., "Extensions of the Spalart-Allmaras Turbulence Model to Account for Wall Roughness," **International Journal of Heat and Fluid Flow**, Vol. 24, 2003, pp. 454-462, https://doi.org/10.1016/S0142-727X(03)00043-2.
- Spalart, P. R., "Trends in Turbulence Treatments," **AIAA 2000-2306**, June 2000, https://doi.org/10.2514/6.2000-2306.

Note that there is a misprint in the AIAA 2000-2306 reference in eq (6). The correct expression for f_{v2} is given below.

The first reference describes two different rough wall methods, one due to Boeing and one due to ONERA. Here, only the method due to Boeing (which appears in both references) is described. A description of the ONERA method, which also includes the use of friction velocity, can be found in the first paper.

The roughness version is the same as for the "standard" version (SA), with the following exceptions. To account for roughness, the distance function, which represents the distance from each field point to the nearest wall, is augmented to read

$$d_{new} = d + 0.03k_s$$

where *d* is the (original) distance to the nearest wall and k_s is the conventional Nikuradse sand roughness scale height. Assuming that k_s is uniform on the body, the new distance definition is used to replace all occurrences of *d* in the original model. The definition of χ is modified to be:

$$\chi = \frac{\hat{\nu}}{\nu} + c_{R1}\frac{k_s}{d_{new}}$$

with $c_{R1} = 0.5$. The new definition of χ should not affect $\hat{S}$, so the definition of f_{v2} needs to be rewritten to read:

$$f_{v2} = 1 - \frac{\hat{\nu}}{\nu + \hat{\nu} f_{v1}}$$

Finally, at the wall (where *d*=0), the boundary condition $\hat{\nu}_{wall} = 0$ is replaced by:

$$\left(\frac{\partial \hat{\nu}}{\partial n}\right)_{wall} = \frac{\hat{\nu}_{wall}}{0.03k_s}$$

where *n* is along the wall normal.

Note that if this roughness model is applied to the (SA-noft2) version instead, then its naming convention becomes (SA-noft2-rough).

Transverse Curvature Free-Shear Correction in Spalart-Allmaras One-Equation Model (SA-TC)

This correction improves the behavior of the SA model in free-shear axisymmetric flows (for example, in axisymmetric jets). The reference is:

- Spalart, P. R. and Garbaruk, A. V., "A New 'Lambda2' Term for the Spalart-Allmaras Turbulence Model, Active in Axisymmetric Flows," **Flow, Turbulence, and Combustion**, Vol. 107, 2021, pp. 245-256; https://doi.org/10.1007/s10494-020-00223-0.

The correction adds the following term to the right-hand side of the SA equation:

$$max(1-r,0)^3 c_{b3}\lambda_2\hat{\nu}/\sigma$$

where c_{b3}=6. The *r* and $\sigma = 2/3$ are the same as in the original model.

The λ_2 is the middle eigenvalue of the Hessian operator of the local SA turbulence variable $\hat{\nu}$. Thus, one must solve the following cubic equation to find the three eigenvalues (keeping the signs):

$$-\lambda^3+(\hat{\nu}_{xx}+\hat{\nu}_{yy}+\hat{\nu}_{zz})\lambda^2+(\hat{\nu}_{xy}^2+\hat{\nu}_{xz}^2+\hat{\nu}_{yz}^2-\hat{\nu}_{xx}\hat{\nu}_{yy}-\hat{\nu}_{xx}\hat{\nu}_{zz}-\hat{\nu}_{yy}\hat{\nu}_{zz})\lambda+(\hat{\nu}_{xx}\hat{\nu}_{yy}\hat{\nu}_{zz}+2\hat{\nu}_{xy}\hat{\nu}_{xz}\hat{\nu}_{yz}-\hat{\nu}_{xy}^2\hat{\nu}_{zz}-\hat{\nu}_{xz}^2\hat{\nu}_{yy}-\hat{\nu}_{yz}^2\hat{\nu}_{xx}) = 0$$

then keep only the middle one, calling it λ_2. One way to find the middle eigenvalue λ_2 is via: sum(L_1+L_2+L_3) - min(L_1,L_2,L_3) - max(L_1,L_2,L_3), where L_i denotes the eigenvalues. The units of λ_2 are the same as the units of the Hessian matrix. Regarding the above notation, note that, for example,

$$\hat{\nu}_{xy} = \frac{\partial^2 \hat{\nu}}{\partial x \partial y}$$

The method for solving a cubic equation can be found at: Wikibooks solution of cubic equations. Also see: Wikipedia Eigenvalue Algorithm.

Spalart-Allmaras One-Equation Model with Quadratic Constitutive Relation, 2000 version (SA-QCR2000)

This nonlinear model version of Spalart-Allmaras is described in:

- Spalart, P. R., "Strategies for Turbulence Modelling and Simulation," **International Journal of Heat and Fluid Flow**, Vol. 21, 2000, pp. 252-263, https://doi.org/10.1016/S0142-727X(00)00007-2.

The model is computed the same as SA, but instead of the traditional linear Boussinesq relation, the following form for the turbulent stress is used:

$$\tau_{ij,QCR} = \tau_{ij} - C_{cr1}\left[O_{ik}\tau_{jk} + O_{jk}\tau_{ik}\right]$$

where τ_{ij} are the turbulent stresses computed from the Boussinesq relation, and O_{ik} is an antisymmetric normalized rotation tensor, defined by:

$$O_{ik} = 2W_{ik}/\sqrt{\frac{\partial u_m}{\partial x_n}\frac{\partial u_m}{\partial x_n}}$$

$$W_{ik} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_k} - \frac{\partial u_k}{\partial x_i}\right)$$

Note that Einstein index notation is used, so that the denominator of the equation for O_{ik} expands to:

$$\sqrt{\frac{\partial u_m}{\partial x_n}\frac{\partial u_m}{\partial x_n}} = \sqrt{u_x^2+u_y^2+u_z^2+v_x^2+v_y^2+v_z^2+w_x^2+w_y^2+w_z^2}$$

In practice, one must avoid division by zero in the O_{ik} term in regions of zero gradient, where QCR should have no effect. The constant in the model is $C_{cr1} = 0.3$. Note that if the QCR2000 correction is added to a different base Spalart-Allmaras model, the naming of the model should reflect it. For example, if adding QCR2000 to (SA-noft2), the new model should be referred to as (SA-noft2-QCR2000). A very common combination is: (SA-RC-QCR2000)

(Note also that the QCR2000 methodology can be used for *any* turbulence models that normally use the Boussinesq relation. When *k* is available, it does not matter whether the $-2\rho k\delta_{ij}/3$ term is included in the expression for τ_{ij} or whether it is added to $\tau_{ij,QCR}$ afterward. The addition is proportional to del_ij, and then the antisymmetry of O_ij makes the two terms cancel.)

Spalart-Allmaras One-Equation Model with Quadratic Constitutive Relation, 2013 version (SA-QCR2013 and SA-QCR2013-V)

The QCR2013 nonlinear model version of Spalart-Allmaras is described in:

- Mani, M., Babcock, D. A., Winkler, C. M., and Spalart, P. R., "Predictions of a Supersonic Turbulent Flow in a Square Duct," AIAA Paper 2013-0860, January 2013, https://doi.org/10.2514/6.2013-860.

This version of QCR is similar to (SA-QCR2000), the only difference being that the modified turbulent stress has an additional term that approximately accounts for the $-2\rho k\delta_{ij}/3$ term in the Boussinesq relation, although only in regions with non-zero strain.

$$\tau_{ij,QCR} = \tau_{ij} - C_{cr1}\left[O_{ik}\tau_{jk} + O_{jk}\tau_{ik}\right] - C_{cr2}\mu_t\sqrt{2S^*_{mn}S^*_{mn}}\delta_{ij}$$

where

$$S^*_{ij} = S_{ij} - \frac{1}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij}$$

and $S_{ij} = (\partial u_i/\partial x_j + \partial u_j/\partial x_i)/2$. The new constant is $C_{cr2} = 2.5$. Note that if the QCR2013 correction is added to a different base Spalart-Allmaras model, the naming of the model should reflect it. For example, if adding QCR2013 to (SA-noft2), the new model should be referred to as (SA-noft2-QCR2013).

Some applications have noted numerical problems with the term $C_{cr2}\mu_t\sqrt{2S^*_{mn}S^*_{mn}}\delta_{ij}$, notably in wake regions where mu_t is not small (see AIAA-2019-0079). A recommended fix for this problem is to use vorticity instead of strain in this term (as documented in Rumsey, C. L., Lee, H. C., and Pulliam, T. H., "Reynolds-Averaged Navier-Stokes Computations of the NASA Juncture Flow Model Using FUN3D and OVERFLOW," AIAA Paper 2020-1304, January 2020). This form is referred to as (SA-QCR2013-V), as follows:

$$\tau_{ij,QCR} = \tau_{ij} - C_{cr1}\left[O_{ik}\tau_{jk} + O_{jk}\tau_{ik}\right] - C_{cr2}\mu_t\sqrt{2W_{mn}W_{mn}}\delta_{ij}$$

(Note that the QCR2013 and QCR2013-V methodologies can be used for *any* turbulence models that normally use the Boussinesq relation. However, if the model provides *k*, then the C_{cr2} term is redundant with the $-2\rho k\delta_{ij}/3$ term in the Boussinesq relation, and the latter term should be used instead.)

Spalart-Allmaras One-Equation Model with Quadratic Constitutive Relation, 2020 version (SA-QCR2020)

The QCR2020 nonlinear model version of Spalart-Allmaras is described in:

- Rumsey, C. L., Carlson, J.-R., Pulliam, T. H., and Spalart, P. R., "Improvements to the Quadratic Constitutive Relation Based on NASA Juncture Flow Data," **AIAA Journal**, Vol. 58, No. 10, 2020, pp. 4374-4384, https://doi.org/10.2514/1.J059683.

This version of QCR uses:

$$\tau_{ij,QCR2020} = \tau_{ij} - C''_{cr1}\left[O_{ik}\tau_{jk} + O_{jk}\tau_{ik}\right] - C''_{cr2}\mu_t\sqrt{2W_{mn}W_{mn}}\delta_{ij}$$

with

$$C''_{cr1} = C'_{cr1}(1 + C_{fw1}f_w)$$

$$C''_{cr2} = C'_{cr2}(1 + C_{fw2}f_w)$$

and $C'_{cr1} = 0.20$, $C'_{cr2} = 1/(3a_1) = 2.15054$, $C_{fw1} = 2.0$, $C_{fw2} = 0.3$, and $a_1 = 0.155$.

The f_w function is originally from Spalart and Allmaras (Recherche Aerospatiale, Vol. 1, 1994, pp. 5-21), later modified in ICCFD7, paper 1902, July 2012 to recover from small negative excursions:

$$f_w = g\left[\frac{1 + c_{w3}^6}{g^6 + c_{w3}^6}\right]^{1/6}$$

where

$$g = r + c_{w2}(r^6 - r)$$

$$r = min\left[\frac{\hat{\nu}}{\hat{S}\kappa^2 d^2}, 10\right]$$

$$\hat{S} = \Omega_s + \bar{S}, for \bar{S} \geq -C_2\Omega_s$$
$$\hat{S} = \Omega_s + \frac{\Omega_s(C_2^2\Omega_s + C_3\bar{S})}{(C_3 - 2C_2)\Omega_s - \bar{S}}, for \bar{S} < -C_2\Omega_s$$

$$\bar{S} = \frac{\hat{\nu}}{\kappa^2 d^2}f_{v2}$$

$$f_{v2} = 1 - \frac{\chi}{1 + \chi f_{v1}}$$

$$f_{v1} = \frac{\chi^3}{\chi^3 + c_{v1}^3}$$

$$\chi = \hat{\nu}/\nu$$

with $c_{w2} = 0.3$, $c_{w3} = 2$, $\kappa = 0.41$, $c_{v1} = 7.1$, $C_2 = 0.7$, and $C_3 = 0.9$.

For QCR2020, a modified function is recommended to replace the vorticity that typically goes into the computation of f_w:

$$\Omega_s = [0.5(2W_{ij}W_{ij} + 2S_{ij}S_{ij})]^{1/2}$$

With this change, converged results are essentially unaffected, but a "smoother" f_w results, which may help numerical convergence behavior.

As with QCR2013 and QCR2013-V, the QCR2020 methodology can be employed for other turbulence models that use the Boussinesq relation. However, if the model provides *k* and it is included in the Boussinesq relation via $-2\rho k\delta_{ij}/3$, then the last term in the $\tau_{ij,QCR2020}$ equation above is redundant and should not be included. If using a non-SA-based model with QCR2020, f_w is not available because $\hat{\nu}$ (in *r*) is not. One simple approximate solution is to use ν_t instead of $\hat{\nu}$ to calculate *r* if $\nu_t/\nu > 10$, and to set *r* = 1 if $\nu_t/\nu \leq 10$ (then use *r* to compute *g* and f_w).

The minimum distance to the nearest wall (*d*) is required for QCR2020.

Spalart-Allmaras One-Equation Model with Quadratic Constitutive Relation, 2024 version (SA-QCR2024)

(This section was contributed by Y. Tamaki of Tohoku University.) The QCR2024 nonlinear model version of Spalart-Allmaras is described in:

- Tamaki, Y. and Kawai, S., "Turbulence Anisotropy Effects on Corner-Flow Separation: Physics and Turbulence Modelling," **Journal of Fluid Mechanics**, Vol. 980 (2024), A21, https://doi.org/10.1017/jfm.2024.25.

This model is shown to improve predictions in a bulging square duct at Re_L=1 million, Re_tau approx 690. Although the original paper shows a slightly different form, the model can be equivalently rewritten into a form similar to QCR2000:

$$\tau_{ij,QCR2024} = \tau_{ij} - C_{cr1}[O_{ik}\tau_{jk} + O_{jk}\tau_{ik}] - C_{cr2}\mu_t \left[S^*_{ik}S^*_{jk} - \frac{1}{3}S^*_{mn}S^*_{mn}\delta_{ij}\right] / \sqrt{\frac{\partial u_m}{\partial x_n}\frac{\partial u_m}{\partial x_n}} - C_{cr3}\mu_t\sqrt{2W_{mn}W_{mn}}\delta_{ij}$$

where O_{ik} and W_{ik} are defined in the SA-QCR2000 section above, and

$$S^*_{ij} = S_{ij} - \frac{1}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij}$$

with $S_{ij} = (\partial u_i/\partial x_j + \partial u_j/\partial x_i)/2$. The constants are: $C_{cr1} = 0.5$, $C_{cr2} = 1.0$, and $C_{cr3} = 2.15$.

As with QCR2013, QCR2013-V, and QCR2020, the QCR2024 methodology can be employed for other turbulence models that use the Boussinesq relation. However, if the model provides *k* and it is included in the Boussinesq relation via $-2\rho k\delta_{ij}/3$, then the last term in the $\tau_{ij,QCR2024}$ equation above is redundant and should not be included.

Spalart-Allmaras One-Equation Model with Velocity Helicity (SA-noft2-Helicity)

*Warning: the additional term in the SA-noft2-Helicity model is **not Galilean invariant**, since it involves the velocity vector. Therefore results will be dependent on your frame of reference. Such a dependence has been avoided in conventional turbulence modeling, and certainly in the original SA model. This lack of Galilean invariance makes this version of the model less general.*

This form of the Spalart-Allmaras model attempts to account for turbulence energy backscatter using velocity helicity. The modification can significantly improve the predictive accuracy for complex 3D vortical flows (such as corner separation in compressors). The reference is:

- Liu, Y.W., Lu, L.P., Fang, L., Gao, F., "Modification of Spalart-Allmaras Model with Consideration of Turbulence Energy Backscatter Using Velocity Helicity," **Physics Letters A**, Vol. 375(24), 2011, pp. 2377-2381, https://doi.org/10.1016/j.physleta.2011.05.023.

This version was implemented by the authors on top of (SA-noft2) (hence the naming convention above). The new feature in this model is that the magnitude of vorticity Ω (in the $\hat{S}$ production term) gets replaced with:

$$(1.0 + C_{h1}h^{C_{h2}})\Omega$$

The final definition of modified $\hat{S}$ is:

$$\hat{S} = \left(1.0 + C_{h1}h^{C_{h2}}\right)\Omega + \frac{\hat{\nu}}{\kappa^2 d^2}f_{v2}$$

where

$$h = \frac{|\omega_i u_i|}{\Omega U} \qquad \omega_i = \epsilon_{ijk}\frac{\partial u_j}{\partial x_k} \qquad U = \sqrt{u_i u_i}$$

The relative helicity density *h* is employed to represent the turbulence energy backscatter. Note: the authors added 0.00001 m/s^2 to the denominator of the equation for *h* to avoid division by zero (private communication). The two constants in the modification are:

$$C_{h1} = 0.71 \qquad C_{h2} = 0.6$$

The modification will switch-off automatically in many classical flows that have been validated for the (SA-noft2) model, because *h* is zero. Although the "Helicity" fix was implemented on top of (SA-noft2), it could also presumably be implemented on top of the general model (SA) or (SA-neg), for example.

A Compressible Form of Spalart-Allmaras One-Equation Model (SA-noft2-Catris)

This particular compressible form was developed by Catris and Aupoix, and is given in the following reference:

- Catris, S. and Aupoix, B., "Density Corrections for Turbulence Models," **Aerospace Science and Technology**, Vol. 4, 2000, pp. 1-11, https://doi.org/10.1016/S1270-9638(00)00112-7.

$$\frac{\partial \rho\hat{\nu}}{\partial t} + u_j\frac{\partial \rho\hat{\nu}}{\partial x_j} = c_{b1}\hat{S}\rho\hat{\nu} - c_{w1}f_w\rho\left(\frac{\hat{\nu}}{d}\right)^2 + \frac{1}{\sigma}\frac{\partial}{\partial x_j}\left(\mu\frac{\partial\hat{\nu}}{\partial x_j}\right) + \frac{1}{\sigma}\frac{\partial}{\partial x_j}\left(\sqrt{\rho}\hat{\nu}\frac{\partial\sqrt{\rho}\hat{\nu}}{\partial x_j}\right) + \frac{c_{b2}}{\sigma}\frac{\partial\sqrt{\rho}\hat{\nu}}{\partial x_i}\frac{\partial\sqrt{\rho}\hat{\nu}}{\partial x_i}$$

There is no trip term, and this model does not include the term f_{t2} from the original model. Because the analysis is restricted to the logarithmic region and, for density gradient effects, to zero pressure gradient flows, Catris and Aupoix do not mention the f_{v1} and f_{v2} terms that appear in the original model (in the production term, in the definition of *r*, and in the definition of turbulent eddy viscosity). However, in the numerical implementation of the model, these should be coded as usual (ref: private communication with the authors). All other functions and constants should be the same as for the "standard" version (SA).

According to Waligura et al. (AIAA-2022-0587, https://doi.org/10.2514/6.2022-0587), this model can be written in a different way that makes it easier to implement:

$$\frac{\partial \rho\hat{\nu}}{\partial t} + \frac{\partial \rho u_j\hat{\nu}}{\partial x_j} - \beta\rho\hat{\nu}\frac{\partial u_j}{\partial x_j} = \rho c_{b1}(1-f_{t2})\hat{S}\hat{\nu} - \rho\left(c_{w1}f_w - \frac{c_{b1}}{\kappa^2}f_{t2}\right)\left[\frac{\hat{\nu}}{d}\right]^2 + \frac{1}{\sigma}\left[\frac{\partial}{\partial x_j}\left(\rho(\nu+\hat{\nu})\frac{\partial\hat{\nu}}{\partial x_j} + \frac{\hat{\nu}^2}{2}\frac{\partial\rho}{\partial x_j}\right) + \rho c_{b2}\frac{\partial\hat{\nu}}{\partial x_i}\frac{\partial\hat{\nu}}{\partial x_i}\right] + \frac{c_{b2}}{\sigma}\left(\hat{\nu}\frac{\partial\rho}{\partial x_i}\frac{\partial\hat{\nu}}{\partial x_i} + \frac{1}{4}\frac{\hat{\nu}^2}{\rho}\frac{\partial\rho}{\partial x_i}\frac{\partial\rho}{\partial x_i}\right)$$

In this equation, the f_{t2} term is included, along with an additional β term to differentiate between the original nonconservative form and a (different) conservative form. When $\beta = 1$ and f_{t2} is set to zero, the original SA-noft2-Catris is recovered. Setting $\beta = 1$ and including f_{t2} would yield SA-Catris. Setting $\beta = 0$ and ignoring f_{t2} yields SA-noft2-CatrisCons. And setting $\beta = 0$ and including f_{t2} yields SA-CatrisCons.

Note that there was a typo in AIAA-2022-0587: the c_{b2} term should be added (as shown above), not subtracted.

Many other papers have been written that either discuss or explore SA-noft2-Catris. A few examples include: AIAA-2016-0586, https://doi.org/10.2514/6.2016-0586, J. Thermophysics and Heat Transfer 2015, 29(2):423-428, https://doi.org/10.2514/1.T3864, Prog. Aer. Sci. 2006, 42:469-530, https://doi.org/10.1016/j.paerosci.2006.12.002, Eng. Applications of Comp. Fluid Mech., 2018, 12(1):459-472, https://doi.org/10.1080/19942060.2018.1451389, and Int. J. Heat and Fluid Flow, 2018, 73:114-123, https://doi.org/10.1016/j.ijheatfluidflow.2018.07.005.

Spalart-Allmaras One-Equation Model with Edwards Modification (SA-noft2-Edwards)

This form was developed primarily to improve the near-wall numerical behavior of the model (i.e., the goal was to improve the convergence behavior). The reference is:

- Edwards, J. R. and Chandra, S. "Comparison of Eddy Viscosity-Transport Turbulence Models for Three-Dimensional, Shock-Separated Flowfields," **AIAA Journal**, Vol. 34, No. 4, 1996, pp. 756-763, https://doi.org/10.2514/3.13137.

This version is the same as for the "standard" version (SA), except that f_{t2} is ignored, and the following two variables are redefined:

$$\hat{S} = \overline{S}^{1/2}\left[\frac{1}{\chi} + f_{v1}\right]$$

$$r = \frac{tanh\left[\hat{\nu}/(\hat{S}\kappa^2 d^2)\right]}{tanh(1.0)}$$

Note that this method makes use of $\overline{S}^{1/2}$ (rather than vorticity Ω), where:

$$\overline{S} = \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\frac{\partial u_i}{\partial x_j} - \frac{2}{3}\left(\frac{\partial u_k}{\partial x_k}\right)^2$$

Spalart-Allmaras One-Equation Model with f_{v3} Term (SA-fv3)

This form of the Spalart-Allmaras model came about as a result of e-mail exchanges between the model developer and early implementers. It was devised to prevent negative values of the source term, and is not recommended because of unusual transition behavior at low Reynolds numbers (see Spalart, P. R., **AIAA 2000-2306**, 2000). Unfortunately, coding of this version still persists. Because this method came about through private communications, there is no official reference for it. However, see the following for a brief description:

- Rumsey, C. L., Allison, D. O., Biedron, R. T., Buning, P.G., Gainer, T. G., Morrison, J. H., Rivers, S. M., Mysko, S. J., and Witkowski, D. P., "CFD Sensitivity Analysis of a Modern Civil Transport Near Buffet-Onset Conditions," NASA/TM-2001-211263, December 2001, https://ntrs.nasa.gov/citations/20020015798.

The equations are the same as for the "standard" version (SA), with the following exceptions:

$$\hat{S} = f_{v3}\Omega + \frac{\hat{\nu}}{\kappa^2 d^2} f_{v2}$$

$$f_{v2} = \frac{1}{(1+\chi/c_{v2})^3}$$

$$f_{v3} = \frac{(1+\chi f_{v1})(1-f_{v2})}{\chi}$$

$$c_{v2} = 5$$

Strain Adaptive Formulation of Spalart-Allmaras One-Equation Model (SA-noft2-salsa)

This form was developed primarily to extend the predictive capabiliy of the model for nonequilibrium conditions. It also makes use of some of the aspects of the (SA-Edwards) version. The reference is:

- Rung, T., Bunge, U., Schatz, M., and Thiele, F., "Restatement of the Spalart-Allmaras Eddy-Viscosity Model in Strain-Adaptive Formulation," **AIAA Journal**, Vol. 41, No. 7, 2003, pp. 1396-1399, https://doi.org/10.2514/2.2089.

This version is the same as for the "standard" version (SA), except for the following changes. First, f_{t2} is ignored. Second, the term

$$\frac{1}{\sigma}\left[\frac{\partial}{\partial x_j}\left((\nu+\hat{\nu})\frac{\partial\hat{\nu}}{\partial x_j}\right)\right]$$

is written slightly differently as:

$$\frac{\partial}{\partial x_j}\left[\left(\nu+\frac{\hat{\nu}}{\sigma}\right)\frac{\partial\hat{\nu}}{\partial x_j}\right]$$

Third, the following two variables are redefined:

$$\hat{S} = S^*\left[\frac{1}{\chi}+f_{v1}\right]$$

$$r = 1.6tanh\left[0.7\sqrt{\frac{\rho_0}{\rho}}\left(\frac{\hat{\nu}}{\hat{S}\kappa^2 d^2}\right)\right]$$

where ρ_0 is the freestream stagnation density, and

$$S^* = \sqrt{2S'_{ij}S'_{ij}}$$

$$S'_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j}+\frac{\partial u_j}{\partial x_i}\right) - \frac{1}{3}\frac{\partial u_k}{\partial x_k}\delta_{ij}$$

Fourth, the sensitization to nonequilibrium effects comes in through a change in c_{b1}, which is no longer a constant. The source term changes from $c_{b1}\hat{S}\hat{\nu}$ to $c'_{b1}\hat{S}\hat{\nu}$ and c_{w1} changes from

$$c_{w1} = \frac{c_{b1}}{\kappa^2} + \frac{1+c_{b2}}{\sigma}$$

to

$$c_{w1} = \frac{c'_{b1}}{\kappa^2} + \frac{1+c_{b2}}{\sigma}$$

where the new variable is

$$c'_{b1} = 0.1355\sqrt{\Gamma}$$

and

$$\Gamma = min[1.25, max(\gamma, 0.75)] \qquad \gamma = max(\alpha_1, \alpha_2)$$

$$\alpha_1 = \left[1.01\left(\frac{\hat{\nu}}{S^*\kappa^2 d^2}\right)\right]^{0.65} \qquad \alpha_2 = max\left[0, 1 - tanh\left(\frac{\chi}{68}\right)\right]^{0.65}$$

Note that in this model the $(2/3)\rho k \delta_{ij}$ term is <u>not</u> ignored in the Boussinesq assumption, and *k* is approximated by

$$k = \nu_t S^* / \sqrt{0.09}$$

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Recent significant updates:
09/06/2024 - added description of QCR2024
03/13/2023 - change to specific recommendation for QCR2020 when used with non-SA-based model
03/01/2023 - clarification about min distance
10/17/2022 - additional clarifications in SA-R
08/16/2022 - clarification on modification to production term in SA-R and SA-KL
02/04/2022 - minor revision on "noft2" designation to SA-Helicity, SA-Catris, SA-Edwards, and SA-salsa, and examples added on how to add corrections
01/24/2021 - added "noft2" designation to SA-Helicity, SA-Catris, SA-Edwards, and SA-salsa to be more accurately descriptive
04/07/2021 - added description of SA-TC
09/15/2020 - added description of QCR2020
07/21/2020 - added description of SA-LRe

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Last Updated: 12/05/2024